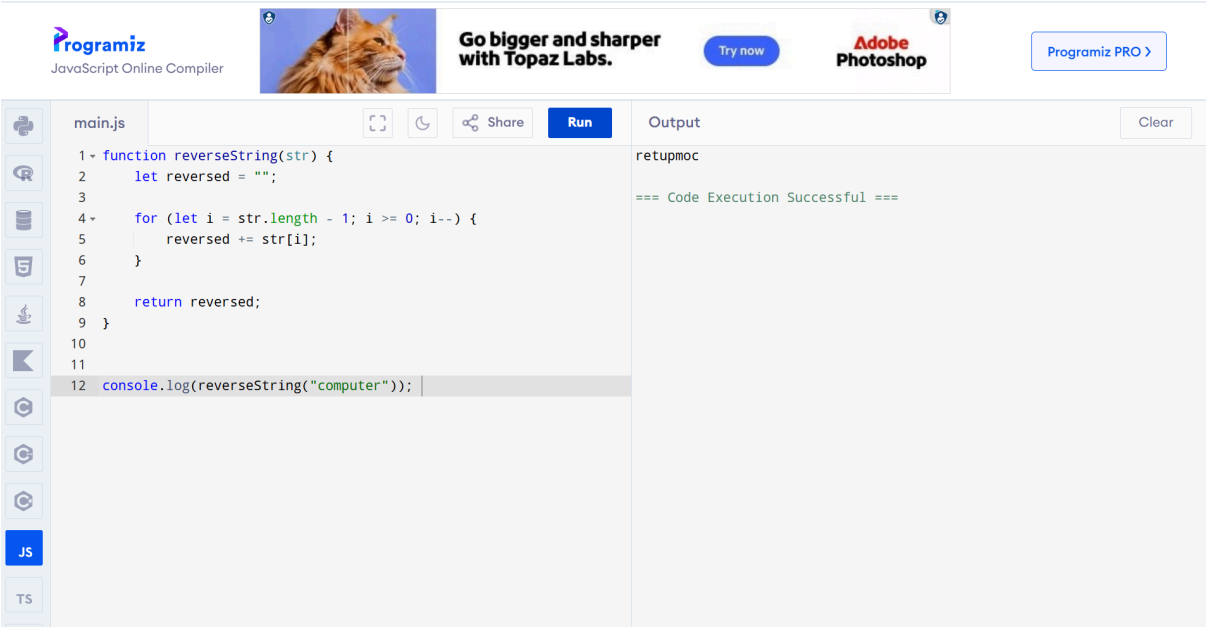
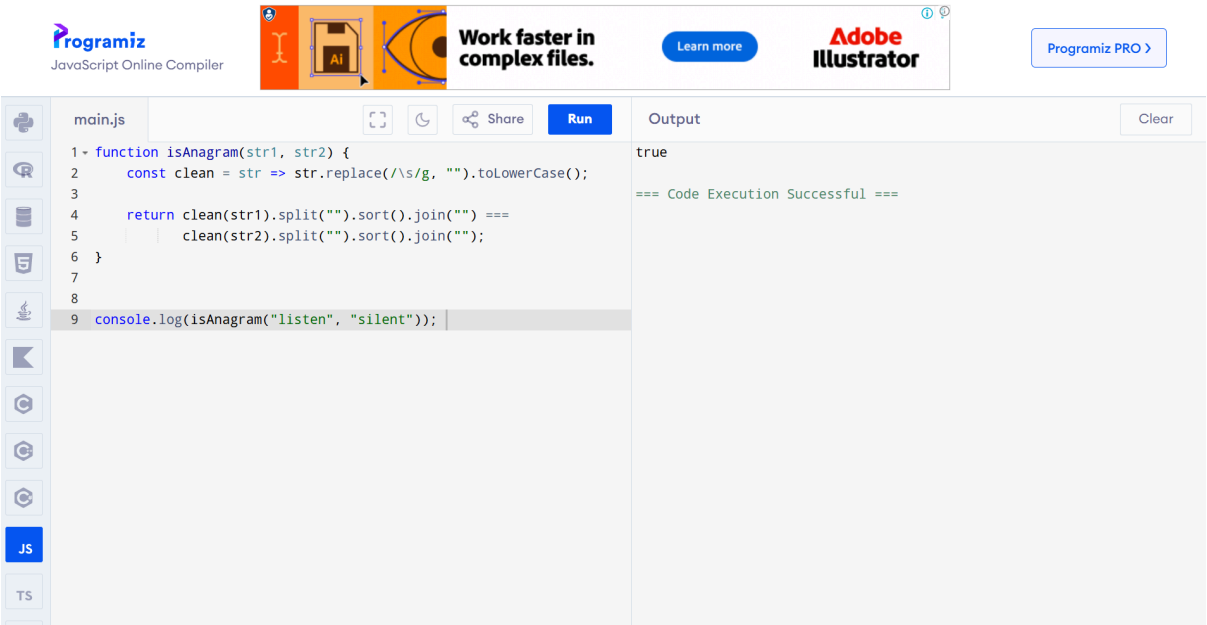
- **String Reversal:** Write a function to reverse a given string in JavaScript without using built-in reverse functions.



- **Anagram Check:** Implement an algorithm to check if two strings are anagrams of each other (contain the same characters with the same frequency)



- **Array Intersection**: Given two arrays, write a function to find their intersection (common elements).

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```javascript
function intersection(arr1, arr2) {
    let set1 = new Set(arr1);
    let result = [];

    for (let num of arr2) {
        if (set1.has(num)) {
            result.push(num);
            set1.delete(num);
        }
    }
  return result;
}
console.log(intersection([1,2,3,4], [3,4,5,6]));
```

```
[ 3, 4 ]

=== Code Execution Successful ===
```

- **String Palindrome:** Create a function to check if a given string is a palindrome (reads the same forwards and backwards) while ignoring non-alphanumeric characters.

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```javascript
function isPalindrome(str) {
    let cleaned = str.replace(/[^a-z0-9]/gi, "").toLowerCase();
    let reversed = "";
    for (let i = cleaned.length - 1; i >= 0; i--) {
        reversed += cleaned[i];
    }
    return cleaned === reversed;
}
```

```
=== Code Execution Successful ===
```

- **Array Rotation:** Implement a function to rotate an array to the right by a specified number of positions.

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main.js

```javascript
function rotateArray(arr, k) {
    let n = arr.length;
    k = k % n;
    let part1 = arr.slice(n - k);
    let part2 = arr.slice(0, n - k);
     return part1.concat(part2);
}
console.log(rotateArray([1,2,3,4,5], 2));
```

Output

```
[ 4, 5, 1, 2, 3 ]

=== Code Execution Successful ===
```

- **String Compression:** Write a function to perform basic string compression using the counts of repeated characters. For example, "aabcccccaaa" would become "a2b1c5a3."

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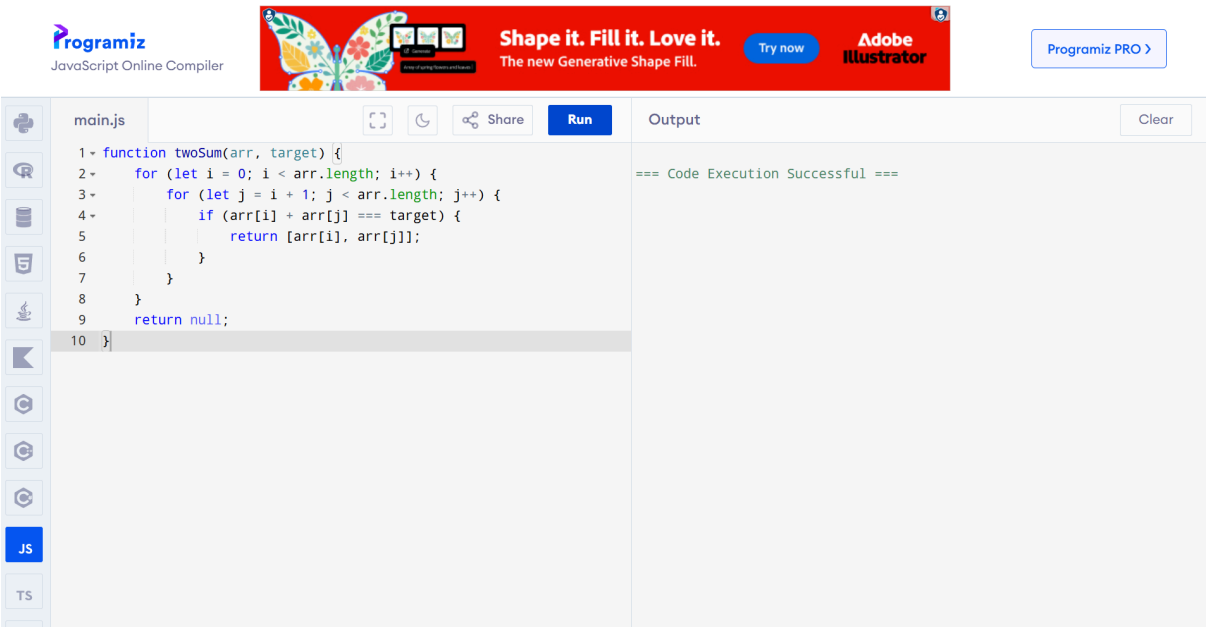
main.js

```javascript
function compressString(str) {
    if (str.length === 0) return "";
    let result = "";
    let count = 1;
     for (let i = 1; i <= str.length; i++) {
        if (str[i] === str[i - 1]) {
            count++;
        }
        else {
            result += str[i - 1] + count;
            count = 1;
        }
    }
 return result;
}
console.log(compressString("aabcccccaaa"));
```

Output

```
a2b1c5a3

=== Code Execution Successful ===
```

- **Array Sum:** Write an algorithm to find the pair of elements in an array that adds up to a specific target sum.